

Lavender Calhoun

+1 (313) 564-9485 | lmcalthoun01@gmail.com | Detroit, MI, USA | LinkedIn | GitHub | Portfolio

EDUCATION

Wayne State University

Bachelor's, Computer Science

Sep 2021 - May 2024

GPA: 3.68

- Cum Laude, 4x Dean's List Student, Society of Computer Developers Member, 6x Vocal Music Grant Awardee

PROFESSIONAL EXPERIENCE

Argonne National Laboratory

Computational Science Intern

Lemont, IL, USA

May 2024 - Aug 2024

- Contributed to a Python library for streamlining machine learning workflows and database queries on Linux systems with high-performance computing nodes.
- Automated dataset generation for molecular dynamics simulations, reducing manual processing by 68%.
- Optimized training cycles by creating workflows to generate datasets addressing model weaknesses.

Wayne State University

Undergraduate Student Researcher

Detroit, MI, USA

Dec 2022 - Apr 2023

- Explored the Neural Radiance Fields (NeRF) algorithm, focusing on its architecture, principles, and applications.
- Refined data preprocessing and model optimization, reducing novel view interpolation error by 15%.

Wilson Adaptive Technologies

Frontend Design Intern

Detroit, MI, USA

Jun 2022 - Aug 2022

- Created prototypes and carried out redesigns for four website pages to enhance user experience, accessibility for customer needs, and branding.
- Developed a comprehensive style guide; standardizing color palettes, fonts, and design elements following the redesign.
- Conducted three user interviews and created detailed profiles to inform design decisions.

MTU Great Lakes Research Center

Data Science and Machine Learning Intern

Houghton, MI, USA

Sep 2020 - Apr 2021

- Developed hands-on experience in machine learning by using the YOLO v3 algorithm to identify armored fighting vehicles from photographic datasets.
- Processed, analyzed, and visualized 5,000+ energy usage data points to identify seasonal trends using Excel, as well as the Pandas, Matplotlib and Seaborn Python libraries.
- Led a presentation on research findings, demonstrating the practical applications of machine learning and data analysis techniques.

PROJECTS & OUTSIDE EXPERIENCE

Tune Tracer

Team Lead/ML Lead

Detroit, MI, USA

Jan 2024 - Apr 2024

- As Team Leader, directed the agile development of a cross-platform app according to client needs and project risks.
- Applied a machine learning model to predict the modality of a song with 94% accuracy.
- As a full-stack developer, designed a record/upload audio function, UI elements in Figma and Dart, and trained the machine learning model.
- Created a backend RESTful API to handle prediction requests using Flask and Heroku.

SKILLS

Languages: Python, C/C++, JavaScript, Java, SQL, Bash, HTML/CSS, Git, Tensorflow, Pytorch, React.js, Docker, MySQL, MongoDB

Skills: Web Development, UI/UX Design, Machine Learning, Software Testing, Software Documentation, Project Management, High Performance Computing (HPC), Human Computer Interaction (HCI)